

Portfolio for Numerai

how to select models

Numerai Comunity Tokyo Meetup 2023 4/8



NUMERAI

habakan (@habakan_f)

Agenda

Implementation of portfolio models in Numerai Tournament Evaluation in live round

- Difficult to select models in local validation because of TC algorithm
 - Unstable TC only single model
- **Methods to stabilize TC with multiple models were verified**

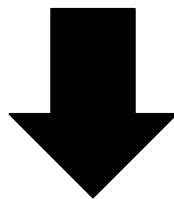
✂Remarks

- **not using V4 Datasets.**
 - I built models still V3 Dataset(2021 release)
- **not validate more models, so low reproducibility**

Why? : bad Data Science UX by TC

**It is difficult to select models using only provided datasets.
Model selection decisions need to be made over a long span of time.**

- Cannot calculation TC using provided datasets.
 - **cannot select models using cross validation previous submission.**
- **Difficulties of built hypothesis and insights about TC in datasets.**



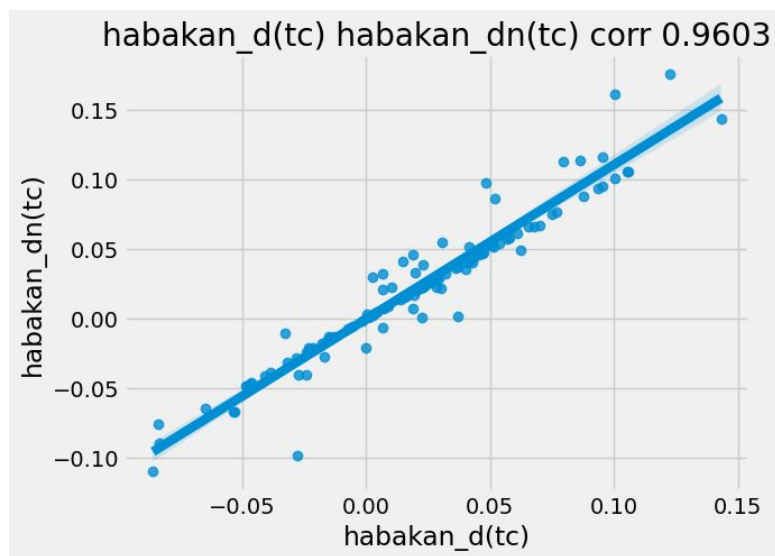
Model selection in Numerai

- TC verification can be done practically during forward testing in Live.
 - **Model selection will be at Stake, not at submission**
- **Model selection in a long cycle PDCA becomes more important**

Which model is good for TC?

habakan_d is a tree base model.
habakan_dn is Feature Neutralization for habakan_d

model	TC sharpe	TC rep	Corr sharpe	Corr rep	All Time Return
habakan_d	0.4474	0.0237	0.6808	0.0176	+ 192.9%
habakan_dn	0.3800	0.0268	0.6975	0.0170	+ 175.6%



- High All Time Return?
- High TC Rep ?
- Both stake?

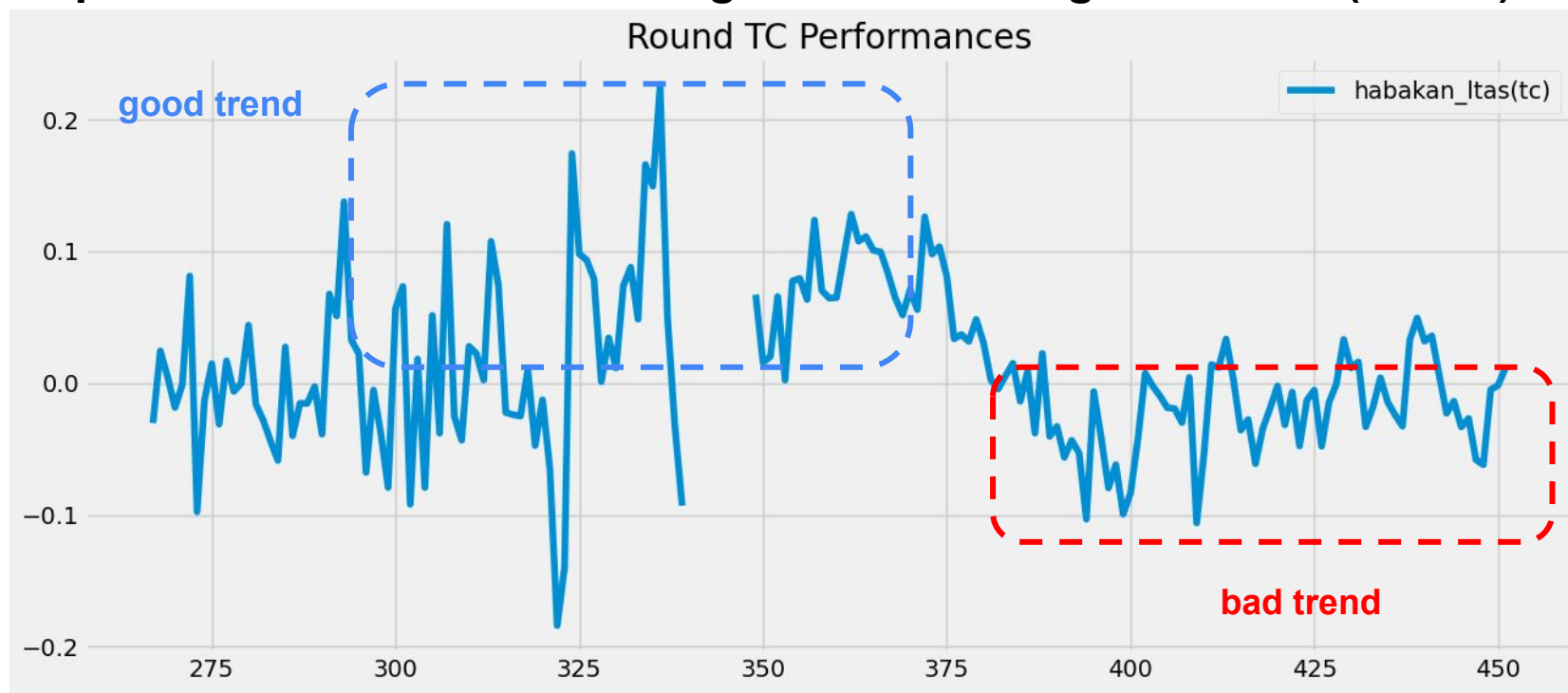
Unstable TC

**Some models have trending TC distribution
Unsure if performance will continue**

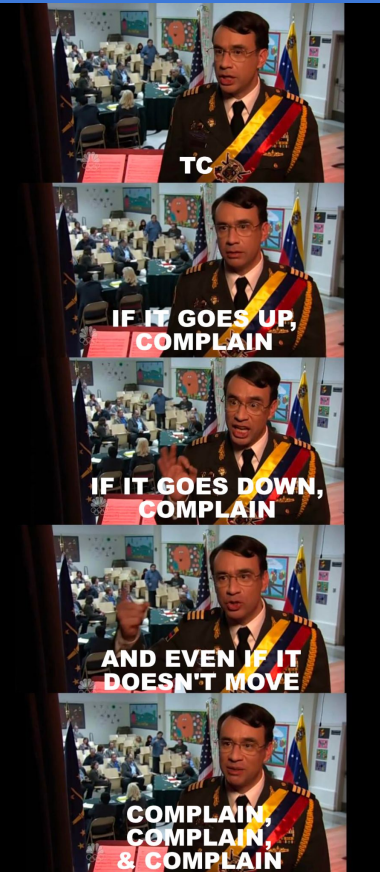
The distribution of TC is not i.i.d because the meta-model is updated.

**→ Even if TC is good at the time, it may become bad
if the distribution changes.**

It is quite difficult to maintain a high TC with a single function (model).



it's not going well for everyone either.



PAYOUTS WILL BE BASED ON TC



katsu1110 @kk1110tt · 2022年2月4日

...えっ...あんなにTwitterでイキってたのに、TCマイナスなの...?



1

↻

9

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よしそ @yosiso44 · 2022年9月16日

TCを上げるために大切なこと。

1. corrを見ない 2. fncを見ない 3. そもそもバックテストしない 4. アイデア勝負 5. データを捨てて直観に従え
データサイエンスとは 😊

1

↻ 9

52

|||

↑



鳩 @pegin_HOLE · 2022年8月17日

レジームチェンジでTCにStakeしたモデルのTCが暴落するのでは...?
globalmacroresearch.org/jp/archives/27...



Very, very strange TC behavior

Tournament

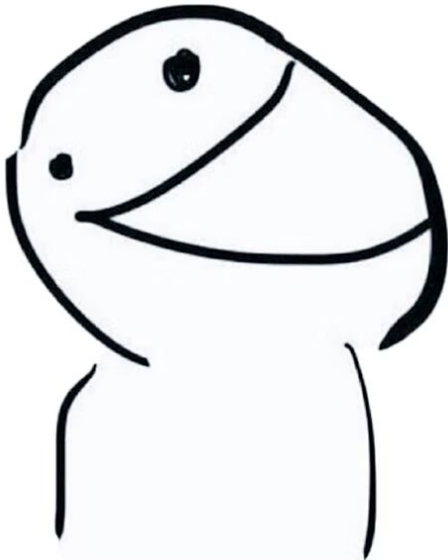
Is TC slowing down your research and experimentation?

Data Science

Do really someone feel comfortable with TC?

Tournament

**If one model is not stable,
Let's Stake multiple models.**



Consider operating with multiple models

**Operate multiple models and optimization portfolio
but the portfolio is unstable and unconvincing**

Challenges to Portfolio Optimization

**Discomfort with optimization
using non-stationary variables
such as TC**

**Observation round is different for
each model.**

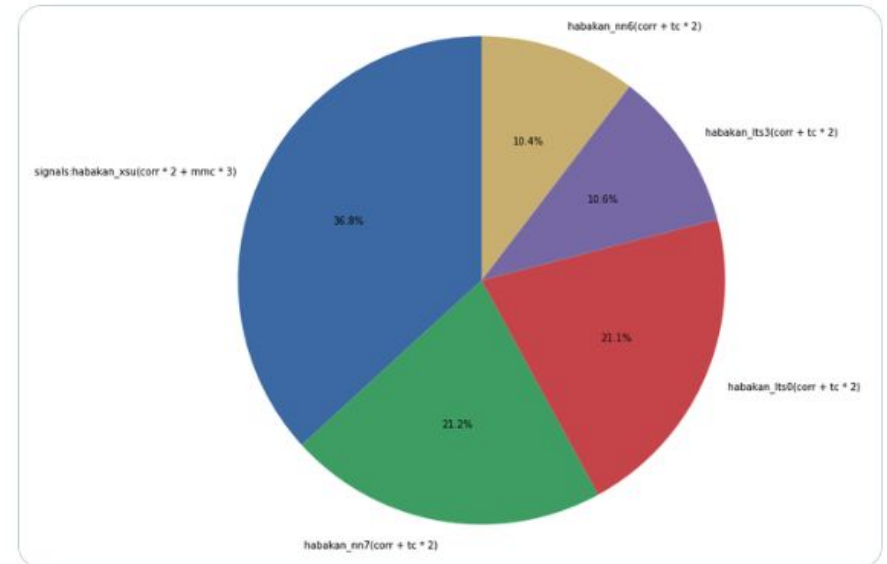
**Correlation between models is
high after all.**



habakan
@habakan_f

...

Stakeポートフォリオが大事になってきそうなので、CVaRとかで最適化するコードを雑に書いた
メイン・Signalsともに自分のLBトップモデルがなくなっていて、さすがに実装ミスを疑っている



午後7:22 · 2022年4月8日



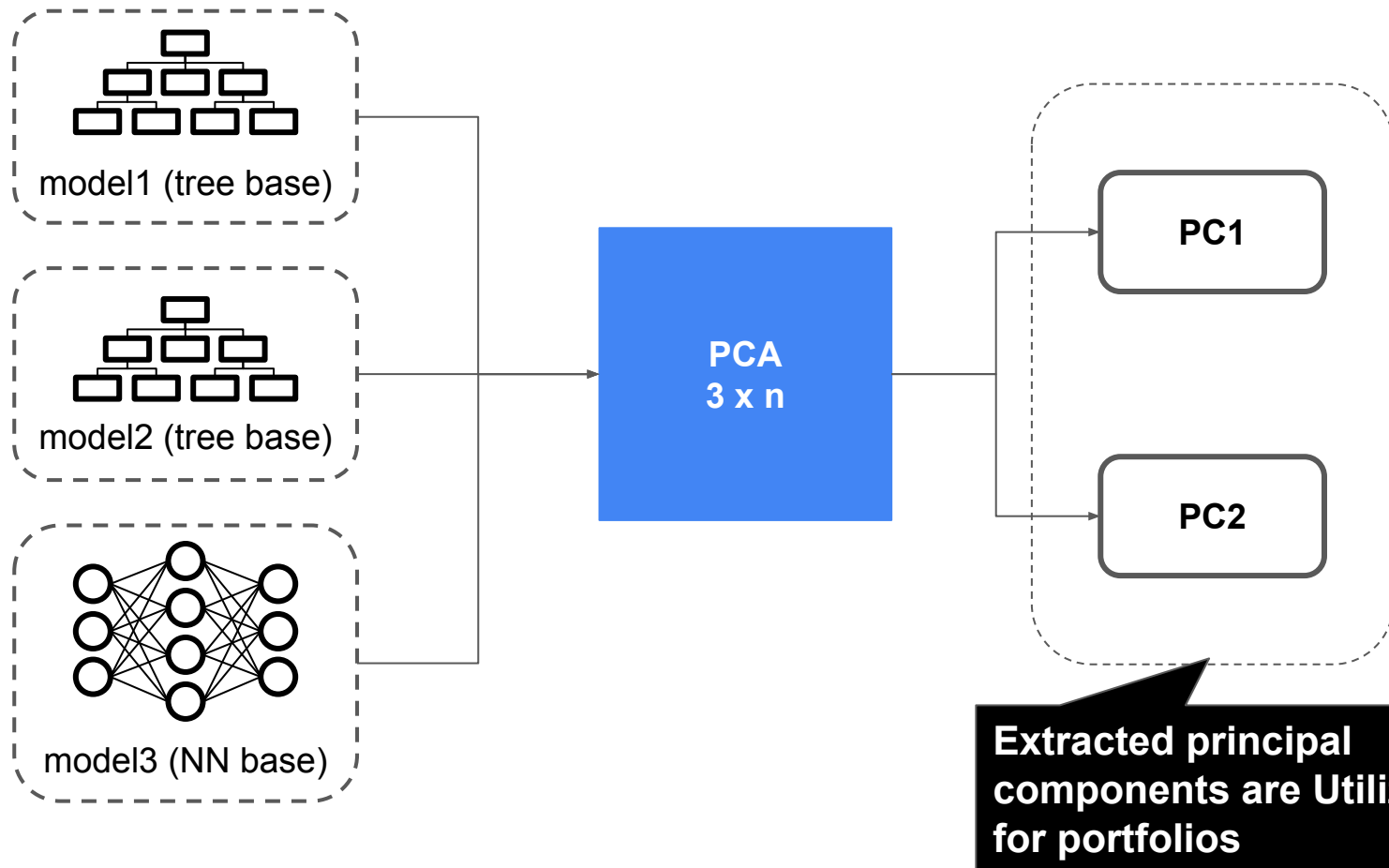
**To expect from portfolio optimization is
to create combinations with small covariance**

The diagram consists of two rounded rectangular boxes with blue borders. The top box contains the text 'To expect from portfolio optimization is to create combinations with small covariance'. A vertical line with an upward-pointing arrow at the top and a downward-pointing arrow at the bottom passes through the center of both boxes. A large, solid black downward-pointing arrow is positioned between the two boxes. From the bottom of the vertical line, a horizontal arrow points to the right, and a diagonal arrow points down and to the left.

**Wouldn't it be better if the model predictions
themselves were each orthogonal?**

Built portfolio using PCA

**Extract principal components from base models,
and add principal components to portfolio**



Built portfolio using PCA

**Extract up to 2nd principal component
from 3 Base models (GBDT x 2 , NN)**

Advantages of using PCA

- **Principal components are orthogonal to each other**
- **Principal components are linear relationship to model predictions**
- **Targets are not necessary and can be extracted unsupervised**

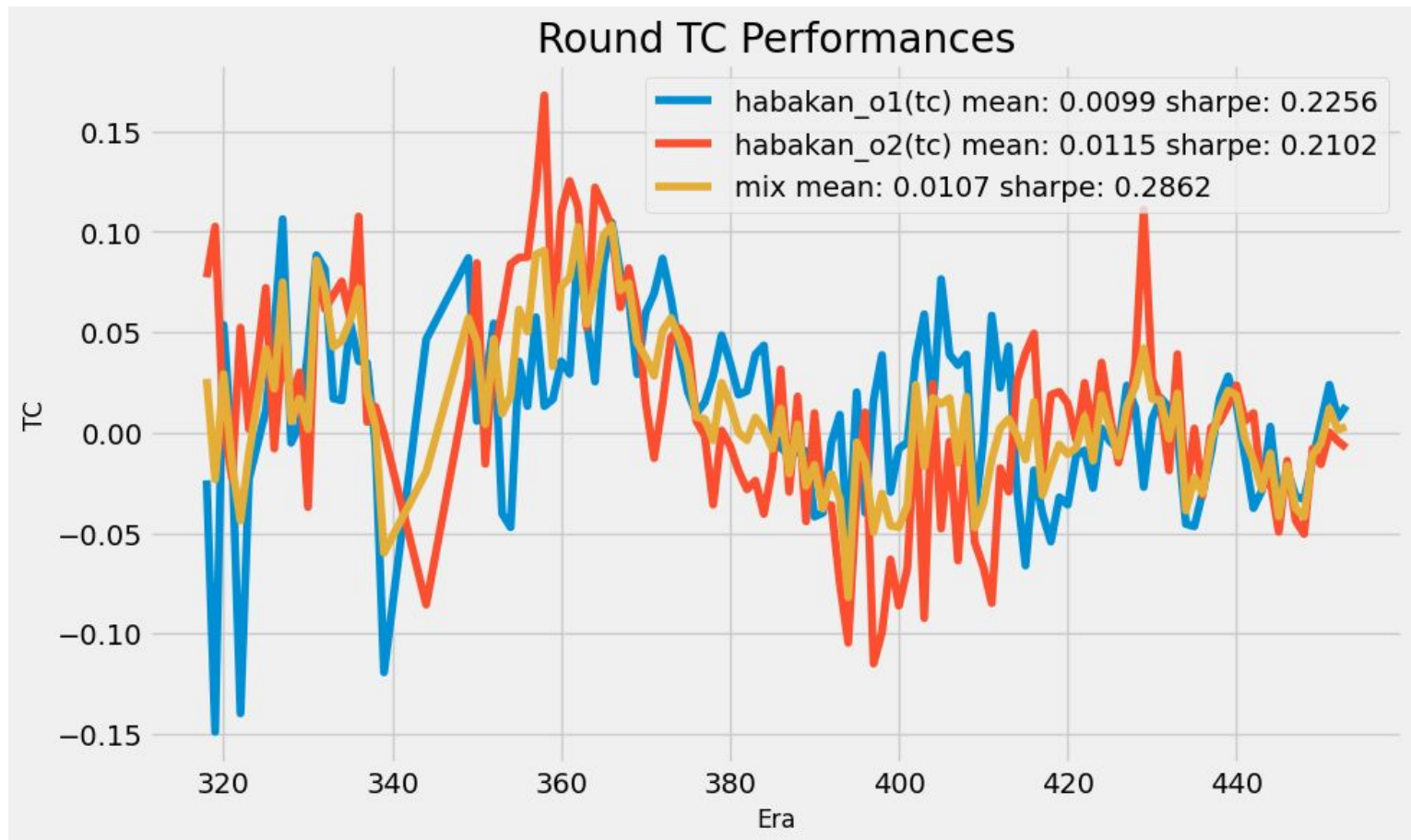
Monitoring and evaluation with LiveRound (318 ~)

- (1) Does TC Sharpe improve if the same amount of principal components are staked?**
- (2) What happens to the TC correlation of the principal components?**
- (3) Does TC Sharpe improve if the same amount of principal components are staked as in the Base model?**

portfolio made up of principal components

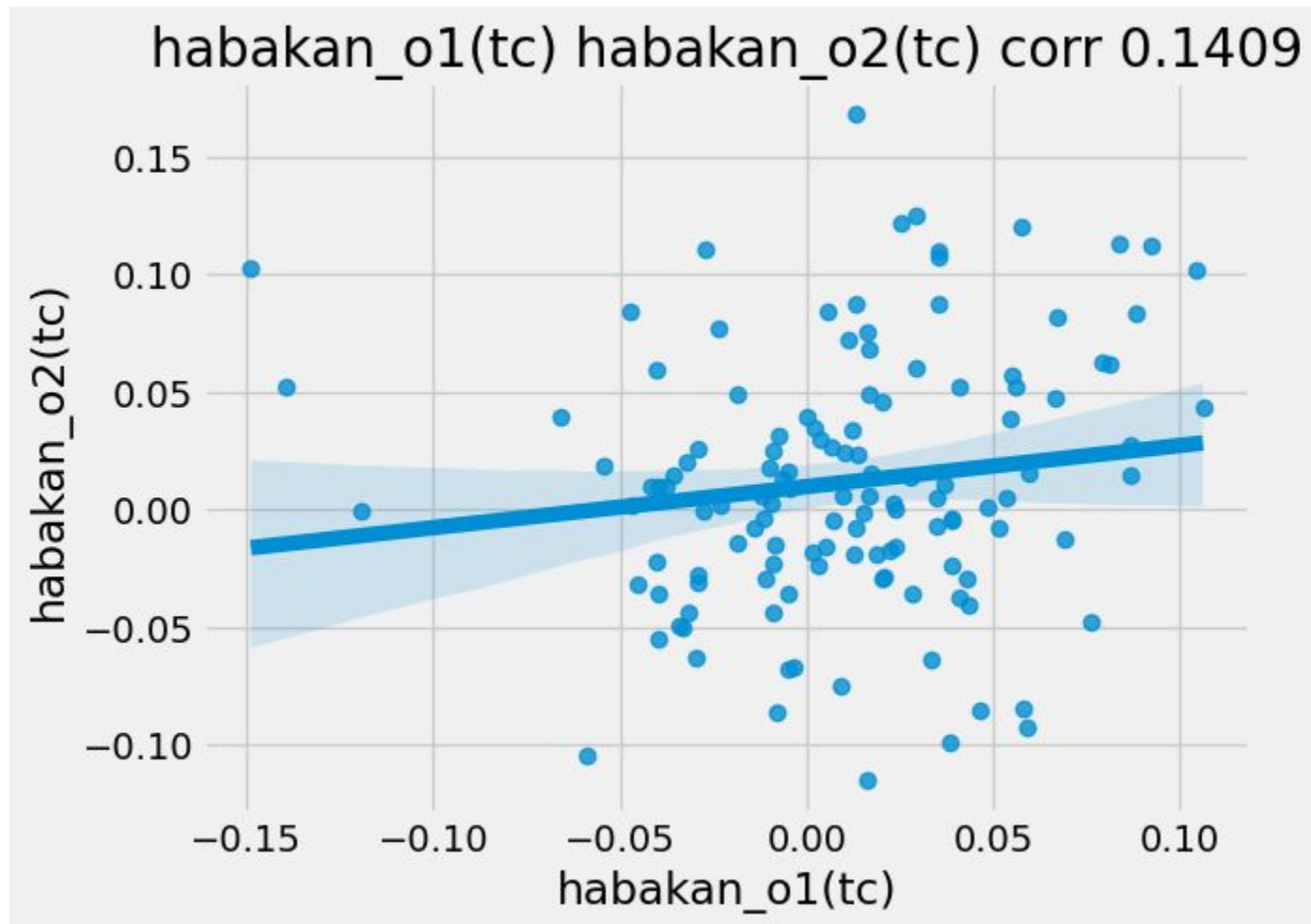
Staking both principal components improves TC Sharpe more than staking one principal component.

- **Stake on both PC1 and PC2 is better than on PC1 and PC2**



portfolio made up of principal components

Mapping TC between principal components
Almost uncorrelated



base model and principal components

Validated with the model with the best TC in the base model and the second principal component

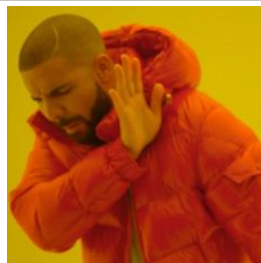
Sharpe is now better in Base model, although it used to improve

Transition of TC Sharpe

	~2022/10	~2022/11	~2022/12	~ today
Base Model1	0.46	0.24	0.44	0.39
PC2	0.97	0.39	0.83	0.22
Base Model1 + PC2	1.03	0.41	1.12	0.37



My TC is up:
"TC is a true
measure of how
creative, unique
and amazing
I am at data science"



My TC is down:
"TC is
impossible to optimize
for, it's
basically gambling
and needs to go away"



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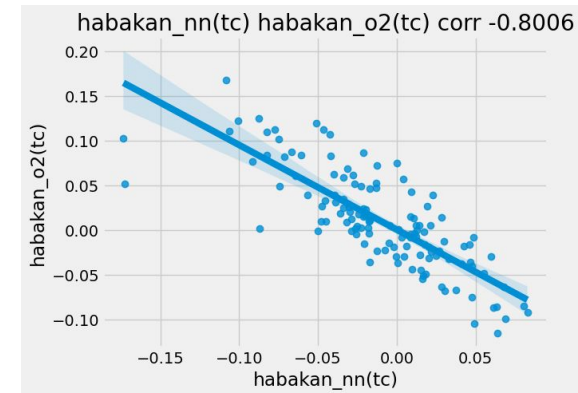
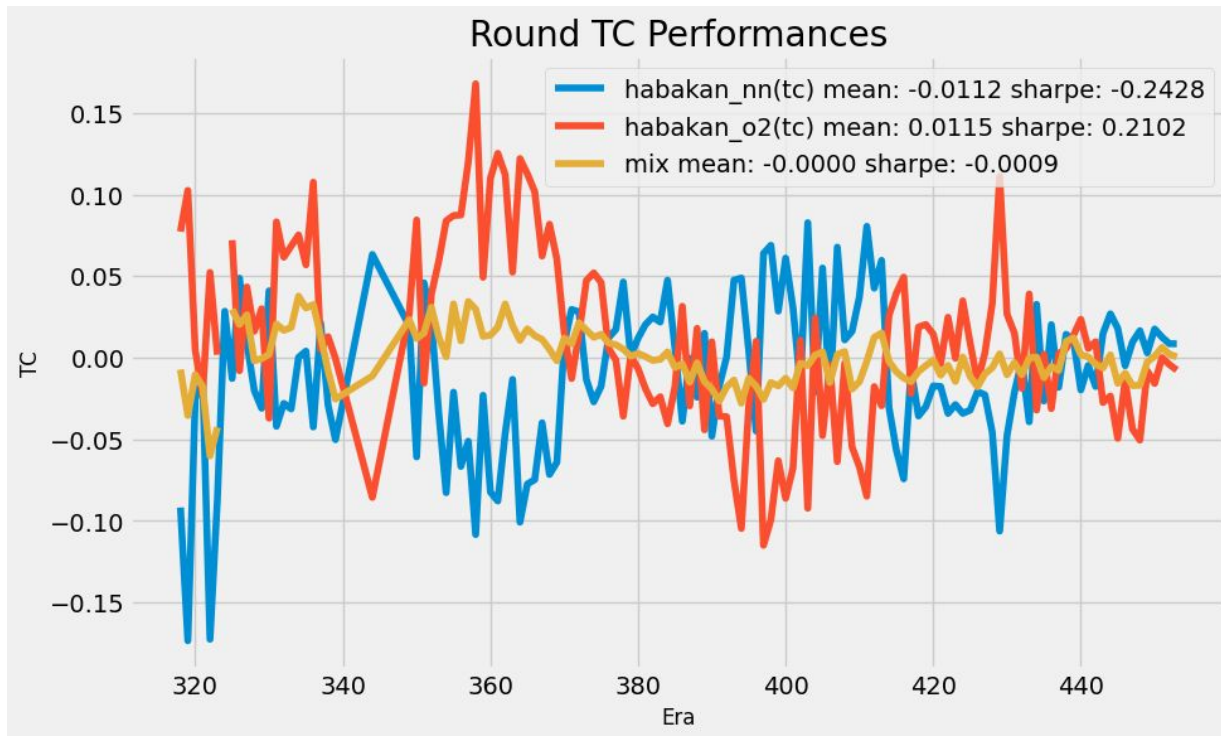


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"TC is
impossible to optimize
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basically gambling
and needs to go away"

Analysis of 2nd principal component

PC2 is almost same as one of base models, NN multiplied by minus
NN has different components from other models
(for this portfolio) It was important that the NN not shift distribution

✂ habakan_nn is worst 377th / 14729th in TC rank



Insights of forwarding test

Principal Component Analysis to uncorrelate TC performance

- **By using PCA, it is possible to construct a portfolio in which TCs are close to uncorrelated.**
- **Sharpe is improved by staking two principal components rather than one.**

Applying PCA does not mean that the Base Model can be outperformed.

- **In the end, it is necessary to select the model to be applied to PCA.**
- **In the original forecasting, Sharpe improved by utilizing the second principal component until the halfway point.**
 - **Sharpe was improved by utilizing the second principal component, but in the end, the result was better with Single only.**
- **However, since we used the same amount of Stake in this case, it may be possible to improve Sharpe by changing the Stake rate.**

**Based on considerations Prospects from each
Stake holder's perspective**

For Numerai participants

Future work for this method

- Even if the result is not good, it can be used as a decision-making tool for model selection.

Currently I am not staking PC2 (NN) because it doesn't improve Sharpe either.

- Select models for PCA

In this case, I selected models that achieved high TC rank.

- Selecting by Lasso using Validation Data
 - Portfolio Optimization after PCA processing
 - Feature Neutralization before PCA
- Are principal components constructed with only nonlinear components valid?
- Can we expect to see Sharpe improvements and a choice of models in the FN & PCA?

For Numerai team

Is there any insight in the analysis of TC scoring based on PCA?

- Does multicollinearity affect TC? (Do TCs converge uniquely?)
- How many ranks does the projected matrix of participants have?
- If I update the gradient of a TC to maximize the return on a given round, what is the vector of true weights? What is the vector of true weights? (Dense? or Sparse?)
- Can't we use PCA itself for TC scoring?
If it would be an easier method of model selection, I feel everyone would be happier if it were incorporated into TC scoring.

For Trader

Are there any trading techniques that take advantage of the orthogonality of PCA?

- Please let me know if anyone has!
I have a feeling that you are doing something similar to applying PCA to a group of factors in factor investing.
- How should we think of multicollinearity in terms of a finance task, which is a difficult task to begin with?
- If we are utilizing machine learning models in finance,
How should we utilize multiple models?
Ensemble ? or Decomposition ?

Let's make matrices, not vectors.

Appendix